BWEM Pilot Study – Scenario 2: Corn Farm Sprinkler Irrigation

This report documents Scenario 2 of the BioLoop Water Efficiency Model (BWEM) pilot simulations. The scenario models a 25-acre corn farm with 30 dairy cattle under semi-arid climatic conditions. Sprinkler irrigation with 75% efficiency was evaluated to estimate applied water and irrigation losses.

1. Methodology

Crop evapotranspiration was calculated using the FAO-56 ETc method (ETc = ETo × Kc). For this scenario, ETo = 4.5 mm/day and Kc = 1.05 for corn. Livestock demand was estimated using CSU Extension water intake guidelines (~35 gallons/day per dairy cow).

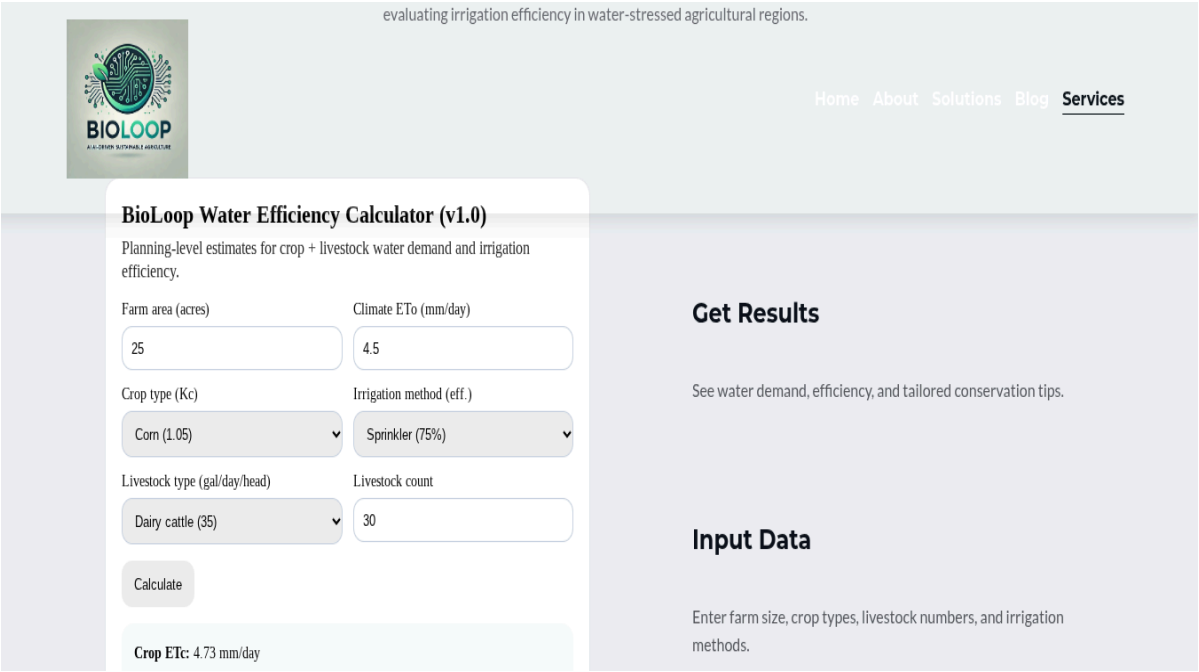
2. Simulation Results

Scenario	Farm Size	Crop	Livestock	Total Demand	Applied Water	Water Loss
Sprinkler (75%)	25 acres	Corn	30 dairy	127,334 gal/day	169,428 gal/day	42,095 gal/day

This scenario demonstrates BWEM's ability to generate demand and loss estimates for a larger acreage farm under sprinkler irrigation.

3. Screenshots from BWEM Simulation

Figure 1. Scenario 2 input and output screenshots



Crop type (Kc)

Irrigation method (eff.)

Sprinkler (75%)

e (gal/day/head)

Livestock count

e (35)

30

Calculate

Crop ETc: 4.73 mm/day
Crop demand: 126284 gal/day
Livestock demand: 1050 gal/day
Total demand: 127334 gal/day
Estimated applied water: 169428 gal/day
Estimated efficiency: 75.2%
Estimated loss: 42095 gal/day

Recommendations:

• Outputs indicate reasonable efficiency; monitor seasonally and adjust as needed.

Note: Planning-level estimates. Local conditions may vary.

See water demand, efficiency, and tailored conservation tips.

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Input Data

Enter farm size, crop types, livestock numbers, and irrigation methods.

Calculate Demand

Estimate water needs based on your specific farm conditions.

How the Water

4. References

Allen, R.G., Pereira, L.S., Raes, D., & Smith, M. (1998). *Crop Evapotranspiration – Guidelines for Computing Crop Water Requirements.* FAO Irrigation and Drainage Paper No. 56. Food and Agriculture Organization of the United Nations.

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Colorado State University Extension. *Calculating Water Requirements for Cattle.* https://goldenplains.extension.colostate.edu/wp-content/uploads/sites/56/2021/06/PR_CalculatingWaterforCattle.pdf

Crop Coefficients for Use in Irrigation Scheduling. https://irrigationtoolbox.com/ReferenceDocuments/Extension/BCExtension/577100-5.pdf

E2 Pilot Study Enhancement — Corn Scenario

Practical Interpretation

The results demonstrate how BWEM can be used to compare irrigation strategies and estimate water demand for corn under different irrigation methods. Differences between irrigation methods can inform decisions regarding water allocation, irrigation efficiency, and resource optimization in semi-arid agricultural settings.

Use Case Scenario

This model can support pre-season agricultural planning by allowing users to estimate crop water requirements for corn and compare irrigation options. It may assist farmers, advisors, and planners in evaluating different irrigation approaches before implementation.

Limitations

This pilot is based on simplified assumptions and is intended for planning-level use only. It does not account for detailed field variability, real-time environmental conditions, or water quality constraints. The model is designed as a decision-support tool rather than a precise operational system.